

Procurement Justification Document

ChipWhisperer-Lite, CW308 UFO Baseboard, and CW308T-STM32F4 Target for TVLA and Embedded Security Research

Purpose: to justify acquisition of a reusable laboratory platform for side-channel leakage assessment, TVLA experiments, embedded cryptography evaluation, and student research training.



Estimated equipment cost: USD 762.50 excluding shipping, tax, and duties

Prepared for: University Research Board / Laboratory Procurement Review

Prepared by: Researcher / PhD Candidate

Date: 29 June 2026

1. Executive Summary

This document requests approval to purchase a ChipWhisperer-based measurement platform consisting of the ChipWhisperer-Lite Arm / 32-bit starter kit, the CW308 UFO target baseboard, and the CW308T-STM32F4 target board with hardware cryptography support. The requested items form a complete, reusable laboratory setup for practical embedded security research, especially power side-channel analysis and Test Vector Leakage Assessment (TVLA).

The main purpose of the acquisition is to support research on cryptographic implementations running on real embedded microcontrollers. In my current research context, the immediate use case is to evaluate leakage behavior of cryptographic code and countermeasures, including post-quantum cryptography implementations such as ML-KEM, on an Arm Cortex-M class target. However, the same platform is not limited to one project. It can be reused by other students working on embedded systems, hardware security, IoT device security, firmware security, lightweight cryptography, random number generation, masking, fault injection, and secure implementation validation.

The three modules should be considered as one integrated research station. The ChipWhisperer-Lite provides capture, triggering, target communication, clocking, and power-measurement capability. The CW308 UFO baseboard provides a flexible and reusable target interface with measurement-friendly power paths, clock selection, programming headers, and compatibility with many interchangeable target boards. The CW308T-STM32F4 target provides a realistic 32-bit STM32F4 microcontroller platform with hardware crypto features, which is highly relevant to modern embedded systems and IoT devices.

2. Requested Modules and Estimated Cost

Module	Manufacturer Part No.	Mouser Part No.	Qty	Unit Price	Role in Research Setup
ChipWhisperer-Lite Arm / 32-bit starter kit	NAE-CWLITE-ARM	343-CWLITEARM	1	USD 368.75	Capture hardware / starter kit for power-analysis workflow
CW308 UFO target baseboard	NAE-CW308-04	343-CW30804	1	USD 343.75	UFO baseboard for mounting target boards
CW308T-STM32F4 target with hardware crypto	NAE-CW308T-STM32F4HWC	343-CW308TSTM32F4HWC	1	USD 50.00	STM32F4 target board for CW308 TVLA / side-channel experiments
Estimated total				USD 762.50	Excludes shipping, duties, taxes, and possible regional price changes.

Note: The price values are taken from the clean Mouser purchase list prepared on 29 June 2026. The final purchase order should verify live Mouser price, stock, regional currency, shipping cost, and import duties before submission.

3. Why This Should Be Treated as a Shared Research Platform

The requested equipment is not a single-purpose component. It is a modular embedded security platform. Once purchased, the laboratory can reuse the capture hardware and CW308 baseboard with different target boards and different student projects.

The CW308 UFO architecture is particularly important because it separates the measurement baseboard from the target microcontroller. This means future students can evaluate other processors, cryptographic accelerators, lightweight crypto devices, IoT-class microcontrollers, or custom target boards without buying a full new capture station each time.

The platform supports both teaching and research. It enables students to move from purely theoretical cryptographic security to practical implementation security, where leakage through power consumption, timing,

clock behavior, and fault behavior can expose secret information even when the mathematical algorithm is secure.

- Side-channel security courses and laboratory exercises.
- TVLA-based leakage assessment for cryptographic software and firmware.
- Evaluation of countermeasures such as masking, hiding, shuffling, randomization, and constant-time design.
- Embedded firmware research using Arm Cortex-M targets.
- IoT security experiments involving real microcontroller hardware.
- Fault-injection and glitching demonstrations, subject to safe laboratory procedure and supervision.
- Benchmarking secure implementations under different compiler settings, clock settings, and target configurations.
- Student projects comparing software crypto, hardware crypto, and post-quantum crypto implementations.

4. Role of Each Requested Module

4.1 ChipWhisperer-Lite Arm / 32-bit Starter Kit (NAE-CWLITE-ARM)

	This is the main capture and control instrument. It provides the measurement interface needed to collect power traces, coordinate trigger signals, communicate with the target, and run repeatable side-channel experiments. In a TVLA workflow, repeatability is essential because the researcher must collect many traces under controlled fixed-versus-random input conditions and then statistically compare the populations.
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Why it is needed

- It provides a dedicated, research-grade capture interface instead of relying on a general oscilloscope-only workflow.
- It integrates capture, trigger, target communication, and software control in the ChipWhisperer ecosystem.
- It allows repeatable experiments where firmware input, capture timing, and trace storage can be automated from Python notebooks or scripts.
- It reduces the time required for new students to start side-channel experiments because many tutorials, examples, and APIs already target this hardware family.

Role in my research

- Capture power traces while cryptographic code runs on the STM32F4 target.
- Automate fixed-versus-random test input collection for TVLA.
- Provide a common baseline capture platform so experimental results can be reproduced and compared.
- Support future testing of implementation-level leakage in ML-KEM, AES, masking experiments, and secure firmware routines.

Value for other students

- Students can use the same capture device for side-channel labs, embedded security projects, cryptography implementation analysis, and introductory hardware security training.
- The device can be reused with multiple targets and not only with the STM32F4 board requested in this purchase.

4.2 CW308 UFO Target Baseboard (NAE-CW308-04)

	The CW308 UFO is the reusable target mainboard. It sits between the capture instrument and interchangeable target boards. Its main value is that it makes the target side of the experiment modular, measurable, and easier to configure. Instead of designing a new measurement PCB for every microcontroller, the laboratory can reuse the CW308 baseboard and install different compatible target boards.
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Why it is needed

- It provides a stable and measurement-oriented base for external targets.
- It supports interchangeable target boards, which protects the university investment over multiple projects.
- It provides useful infrastructure for target experiments, including power regulation, clock-related configuration, and standard programming/debug headers.
- It enables experiments on a wider range of targets than the fixed target section of a simple starter board.

Role in my research

- Hold and interface the STM32F4 target board in a controlled way.
- Support cleaner power measurement paths and repeatable setup for trace collection.
- Allow future replacement of the STM32F4 target with other CW308-compatible targets if research expands to other microcontrollers or hardware accelerators.

Value for other students

- Other students can reuse the same baseboard with different target boards for AES, lightweight crypto, PQC, firmware leakage, IoT device, and fault-injection studies.
- The baseboard reduces future procurement cost because the lab does not need a separate complete platform for each target family.

4.3 CW308T-STM32F4 Target with Hardware Crypto (NAE-CW308T-STM32F4HWC)

	The STM32F4 target board is the device under test. It provides a realistic 32-bit Arm Cortex-M microcontroller environment that is close to devices commonly used in embedded systems, IoT products, industrial controllers, and secure firmware development. The hardware crypto variant is especially valuable because it allows comparison between software cryptographic implementations and microcontroller-supported cryptographic operations.
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Why it is needed

- The research requires a real microcontroller target, not only simulation.
- STM32F4 is a widely used embedded platform, so results are more relevant to practical embedded systems than toy targets.
- The board is compatible with the CW308 UFO baseboard, which keeps the setup modular and reusable.
- Hardware crypto support enables experiments comparing software implementation leakage and hardware-accelerated cryptographic operations.

Role in my research

- Run the cryptographic firmware that will be evaluated using TVLA and side-channel measurement.
- Provide a realistic Arm Cortex-M platform for implementing and benchmarking ML-KEM-related routines, AES baselines, masking experiments, and countermeasure code.

- Allow the research to study leakage behavior under realistic firmware build settings, clocking conditions, and embedded constraints.

Value for other students

- Students working on embedded firmware, IoT security, secure boot, cryptographic libraries, and hardware security can use the STM32F4 target as a common experimental board.
- The same target can support introductory experiments and advanced research because it is simple enough to program but realistic enough for research-quality measurements.

5. Why All Three Modules Are Needed Together

1. Capture and Control ChipWhisperer-Lite Collects traces, controls trigger and communication, stores measurements.	2. Reusable Target Infrastructure CW308 UFO baseboard Provides measurement-friendly target mounting, power/clock/programming infrastructure, and modularity.	3. Device Under Test CW308T-STM32F4 Runs the firmware and cryptographic implementation being evaluated.
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- Buying only the capture board would not provide the reusable external target infrastructure required for clean CW308-based experiments.
- Buying only the CW308 UFO and STM32F4 target would not provide the integrated capture and control hardware needed to collect repeatable side-channel traces.
- Buying only the STM32F4 target would not create a working side-channel measurement setup. It is the device under test, but it needs both the baseboard and the capture instrument to become useful for TVLA work.
- Together, the three items create a complete measurement chain: target firmware execution, controlled triggering and communication, power-trace acquisition, and statistical leakage analysis.

6. Use in TVLA and Side-Channel Research Workflow

The requested setup supports the following experimental workflow:

1. Implement or port the cryptographic routine to the STM32F4 target firmware.
2. Configure fixed-versus-random or other selected test-vector groups according to the leakage assessment plan.
3. Use ChipWhisperer-Lite to communicate with the target and collect a large number of aligned power traces.
4. Use statistical testing such as TVLA to check whether the measured traces show data-dependent leakage.
5. If leakage is detected, modify the firmware or countermeasure design and repeat the experiment under the same measurement configuration.
6. Document results with trace counts, test parameters, compiler settings, clock settings, firmware version, and statistical output.

Research impact: This workflow turns implementation security into an empirical process. It allows the research group to move beyond claiming that an algorithm is secure and instead evaluate whether a real embedded implementation leaks information under measurable physical behavior.

7. Proposed Laboratory Use and Governance

- Keep the equipment under laboratory inventory as a shared embedded security measurement station.
- Require users to book the platform through the supervisor or lab manager to avoid damage and conflicting experiments.
- Maintain a simple experiment logbook including project name, firmware version, target board, measurement configuration, and date.
- Store reusable scripts, notebooks, firmware templates, and setup notes in a shared university repository.
- Train new users with a short safety and setup checklist before independent use.
- Preserve cables, jumpers, target boards, and accessories together in one labeled storage box.

8. Risk, Limitation, and Mitigation

Risk / Limitation	Impact	Mitigation
Price and stock may change before purchase	Final cost may differ from this estimate	Verify live Mouser pricing and stock immediately before purchase order submission.
Improper handling may damage target boards	Loss of equipment availability	Use supervised training, ESD precautions, labeled cables, and experiment logbook.
Students may need time to learn the toolchain	Initial ramp-up time	Use existing ChipWhisperer tutorials and create local lab notes for repeatable setup.
One station may become a bottleneck if many students use it	Scheduling conflicts	Use booking policy; add additional target boards later if demand grows.

9. Recommendation

I recommend purchasing the three requested modules as a single integrated platform. The acquisition is justified because it directly supports the current TVLA and embedded cryptography research while also creating reusable laboratory infrastructure for future student projects.

The total estimated cost is modest compared with building a custom measurement setup from separate instruments and custom target boards. The modular design also protects the university investment because the capture hardware and CW308 UFO baseboard can be reused with other targets in future research.

Approving this purchase will help establish a practical embedded security capability in the laboratory and will support stronger research outputs in cryptographic implementation security, IoT security, firmware security, and hardware security education.

10. Technical Sources and Product References

- **ChipWhisperer-Lite documentation:** <https://chipwhisperer.readthedocs.io/en/latest/Starter%20Kits/ChipWhisperer-Lite.html>
- **ChipWhisperer-Lite capture documentation:** <https://chipwhisperer.readthedocs.io/en/latest/Capture/ChipWhisperer-Lite.html>
- **CW308 UFO documentation:** <https://chipwhisperer.readthedocs.io/en/latest/Targets/CW308%20UFO.html>
- **ChipWhisperer targets overview:** <https://chipwhisperer.readthedocs.io/en/latest/Targets/index.html>
- **CW308T-STM32F documentation:** https://chipwhisperer.readthedocs.io/en/latest/chipwhisperer-target-cw308t/CW308T_STM32F/README.html
- **NewAE CW308T-STM32F hardware crypto documentation:** <https://rtfm.newae.com/Targets/UFO%20Targets/CW308T-STM32F/>
- **Mouser NAE-CWLITE-ARM product page:** <https://www.mouser.com/ProductDetail/NewAE/NAE-CWLITE-ARM?qs=r5DSvlrkXmIUEZprO%2Fombw%3D%3D>
- **Mouser NAE-CW308-04 product page:** <https://www.mouser.com/ProductDetail/NewAE/NAE-CW308-04?qs=r5DSvlrkXmKNVdAQ9%252BKpFQ%3D%3D>
- **Mouser NAE-CW308T-STM32F4HWC product page:** <https://www.mouser.com/ProductDetail/NewAE/NAE-CW308T-STM32F4HWC?qs=r5DSvlrkXmLrgVVFt%252BpyiQ%3D%3D>

Appendix A. Purchase Links

Item	Mfr. Part No.	Mouser Part No.	Mouser Link
ChipWhisperer-Lite Arm / 32-bit starter kit	NAE-CWLITE-ARM	343-CWLITEARM	Open Mouser page
CW308 UFO target baseboard	NAE-CW308-04	343-CW30804	Open Mouser page
CW308T-STM32F4 target with hardware crypto	NAE-CW308T-STM32F4HWC	343-CW308TSTM32F4HWC	Open Mouser page